CREST AWARDS | GOLD

This CREST Gold Award is presented to:

Aarav Jain

for the project

Smart Irrigation System

13/06/2025

Step by Step School

Heather King

Dr Heather King
Vice-President for Education, British Science Association

Managed by BRITISH SCIENCE ASSOCIATION Supported by UK Research and Innovation Urenco

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Knowledge Prism

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To Whomsoever It May Concern

Date: 7th July, 2026

As the Director of the Indigo Knowledge Prism Foundation and Founder of Learnovate, I have spent years working at the intersection of education, innovation, and community development. Through curriculum initiatives, experiential learning programs, and grassroots technology projects, I have had the privilege of mentoring and evaluating hundreds of young students. It is therefore with great enthusiasm that I recommend Aarav A. Jain, a remarkable young innovator whose rare combination of technical curiosity, community leadership, and social consciousness distinguishes him far beyond his peers.

I first came to know Aarav through LinkedIn. Unlike many students who reach out seeking short-term volunteering opportunities, Aarav approached me with a clear purpose. He spoke about his longstanding fascination with water systems, shaped in part by years spent as a swimmer and by his growing concern about the sustainability challenges surrounding water use in India. What particularly stood out was that his interest extended beyond environmental rhetoric. He was actively seeking to understand how technology could be deployed to address real-world problems.

Our conversation revealed a young man who had already begun this journey. During a visit to agricultural communities in Assam, Aarav had spent time observing irrigation practices and interacting with farmers. What he witnessed left a lasting impression. Despite increasing concerns around water scarcity, large volumes of water were still being lost through inefficient irrigation systems and manual monitoring practices. Rather than treating this as an isolated observation, Aarav began investigating whether affordable automation could help farmers conserve water while reducing the burden of daily irrigation management. That inquiry ultimately evolved into the concept of a low-cost smart irrigation system.

Around the same time, Aarav had established himself as a committed mentor and educator. Whether guiding younger students within his school community, supporting participants through STEM outreach programs, or teaching underserved children through educational initiatives, he consistently demonstrated a belief that technology becomes most meaningful when it empowers others. This philosophy naturally aligned with the mission of our Village Research Lab initiative, which brings together students from rural communities to solve local challenges using emerging technologies.



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Over a period of 2 weeks, Aarav worked closely with 15 students from farm families in a village located approximately 30 kilometers outside Gurugram. Rather than building the irrigation system himself and presenting it as a finished product, Aarav chose a far more challenging approach: teaching the students to design, assemble, and deploy the systems themselves.

Under his guidance, the students learned to integrate soil moisture sensors with ESP32 microcontrollers, relay-controlled pumps, and OLED displays to create autonomous irrigation systems capable of monitoring soil conditions and regulating water usage in real time. What impressed me most was not the sophistication of the technology, but Aarav's approach to teaching. He consistently encouraged students to investigate, experiment, and troubleshoot independently rather than relying on him for answers. In doing so, he transformed technical instruction into genuine problem-solving.

As the project evolved, Aarav continually pushed the students to think beyond engineering specifications and consider broader questions of impact. He encouraged discussions around water conservation and long-term sustainability. His ability to connect technical design with practical implementation reflected a maturity that is rarely seen at his age.

The results were significant. By the conclusion of the program, the students had successfully deployed smart irrigation systems across 15 family-owned farms. Early observations indicated approximately a 30% reduction in water consumption compared to conventional irrigation practices. Equally important was the confidence the students gained through the process. Many began viewing technology not as something distant or inaccessible, but as a tool they could use to improve conditions within their own communities.

Over the years, I have worked with many talented young people, but very few have demonstrated Aarav's combination of intellectual curiosity, leadership, and commitment to service. He brings seriousness to his work without losing humility, and ambition without losing sight of the communities he hopes to serve. I am confident that wherever Aarav chooses to study and contribute next, he will continue developing solutions that are both innovative and meaningful. He is the kind of individual who does not wait for opportunities to create impact—he actively seeks them out and follows through with exceptional dedication.

I wholeheartedly recommend Aarav A. Jain with my highest enthusiasm.

Sincerely,
Sachin Kumar
Director
Indigo Knowledge Prism Foundation

SYMBOL OF QUALITY

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To Whom It May Concern,

I am pleased to recommend Aarav Jain, a student of Step-by-Step School, Noida, whose initiative and practical approach to engineering impressed us at Rungta Irrigation. Aarav first reached us through a cold email, attaching a roadmap for constructing a smart irrigation device and proposing that we consider its potential for wider adoption. For a company that designs and supplies irrigation systems, it was an intriguing introduction: a school student was offering technology already deployed across 35 farms, along with a practical plan for building it. That combination of initiative and preparation opened our conversation.

Aarav had developed the device after observing farmers spending hours watering fields manually, often without knowing the optimum level of water needed by the crops, often leading to under or over watering. His system uses solar-powered ESP32-C3 nodes to monitor soil moisture and activate irrigation when readings fall below a set threshold. An on-site display and mobile monitoring through Blynk give farmers visibility and control. As he explained the design, he consistently connected its technical features to everyday concerns: saving water, reducing manual effort, and keeping the system manageable for its users. He also brought evidence-controlled testing recorded a 30% reduction in water use compared with manual irrigation. Working with representatives of the Government of Assam, he deployed the system across 20 farms in Titabor, where farmers reported reduced water consumption and better crop yield.

What made our discussions memorable was how readily Aarav moved from circuitry to commercial questions. What could a smallholder farmer realistically afford? Would the water and labour savings justify the purchase? How could existing distribution and service networks support installations across farms? He recognised that a farmer might reject a technically effective device if repairs were difficult or replacement parts expensive. He examined these concerns as design requirements, showing an understanding of adoption that went well beyond demonstrating a working prototype.

Aarav then made his proposal usable by our team. He shared the intellectual property underlying the device and walked us through an implementable construction roadmap. He trained our employees to assemble the system and explained how its components worked together. This was where his ownership became especially clear: he had arrived independently with a proposal and followed through by giving others the knowledge to build it. Rungta Irrigation purchased his construction roadmap for ₹70,000. We also agreed to advance the system's evaluation for formal adoption within our product catalogue, subject to the necessary technical validation, durability testing, manufacturing assessment, and commercial review.

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From his first email to the technical handover, what stood out was his understanding that innovation has value only when it can move beyond the prototype and create meaningful, sustainable impact in the real world. He thought about the problem as an engineer, and from the perspective of the people who would ultimately use the product: whether it was a user-friendly design, affordable for the intended users, practical to maintain, and capable of being scaled. In doing so, Aarav brought together engineering, human usability, affordability, and commercial thinking with a larger goal in mind—to build something that could be adopted, sustained, and benefit society at scale. I recommend Aarav with confidence. I believe he will bring that practical curiosity, initiative, and sense of responsibility to his university community.

Sincerely,

Shruti Jain

Rungta Irrigation Ltd.